

Project: Predictive Modeling for Real Estate Pricing

Project Objective

Develop a **machine learning model** to predict **house prices** using a synthetic real estate dataset. This project simulates a real-world use case where pricing is influenced by various nonlinear and polynomial relationships among property features.

Tasks & Responsibilities

Phase 1: Data Understanding & Preparation

1. Load & Inspect the Dataset

- Load the provided dataset (`synthetic_real_estate_data.csv`) using pandas.
- Print the shape, column names, and a sample of rows.
- Check data types and ensure they align with expectations.

2. Exploratory Data Analysis (EDA)

- Plot distributions for numeric features: `Area_sqft`, `Age`, `Distance_to_CityCenter_km`.
- Count plot or bar chart for categorical features: `Bedrooms`, `Bathrooms`.
- Visualize target variable (`Price`) distribution.

3. Feature Relationship Analysis

- Use scatter plots to assess relationships between `Price` and features like:
 - `Area_sqft vs Price`
 - `Distance_to_CityCenter_km vs Price`
- Create correlation heatmap to assess linear relationships.

- Identify non-linear patterns visually or using polynomial trendlines.

4. Outlier Detection & Treatment

- Use boxplots and Z-score/IQR method to detect outliers.
- Decide whether to remove, cap, or transform outliers.
- Document any outlier-handling logic clearly.

5. Missing Value Analysis

- Check for missing or null values in the dataset.
- Handle missing values using appropriate strategies (drop, fill, or model-based imputation).
- Justify chosen approach for each affected feature.

6. Data Transformation & Cleaning

- Handle skewed features (e.g., log transform on `Price` or `Distance_to_CityCenter_km` if needed).
- Convert categorical columns to proper types (if needed).
- Ensure data cleanliness (e.g., no negative square footage, proper units).

7. Feature Scaling (if required)

- Assess whether scaling is needed (especially for models sensitive to scale like Ridge, Lasso).
- Apply `StandardScaler` or `MinMaxScaler` to selected features.

8. Documentation & EDA Summary

- Summarize key findings:

- Important patterns or trends
- Features with high correlation to price
- Outlier or missing value treatment decisions
- Save visualizations and data insights for reporting.

Tasks & Responsibilities

Phase 3: Modeling – Build & Evaluate Prediction Model

1. Train-Test Split

- Split the dataset into **training (e.g., 80%)** and **testing (e.g., 20%)** subsets.
- Use `train_test_split` from `sklearn.model_selection`.
- Ensure the target variable (**Price**) is well represented in both sets.

2. Build Baseline Linear Regression Model

- Use `LinearRegression` from `sklearn.linear_model` as a baseline.
- Train on original features without polynomial expansion.
- Evaluate with:
 - RMSE (Root Mean Squared Error)
 - MAE (Mean Absolute Error)
 - R^2 Score

3. Polynomial Feature Expansion

- Use `PolynomialFeatures` from `sklearn.preprocessing` to create polynomial terms:
 - Start with `degree=2`

- Consider interaction-only features as well
- Retrain a **Linear Regression model** on the expanded feature set.
- Compare performance with the baseline model.

4. Save the Best Model

- Use `joblib` or `pickle` to serialize the final model.
- Ensure preprocessing steps (e.g., scaling, polynomial expansion) are saved or pipelined.

5. Build a Full Modeling Pipeline (Optional but Ideal)

- Use `Pipeline` from `sklearn.pipeline` to combine:
 - Feature preprocessing
 - Polynomial expansion
 - Regression model
- Makes model deployment and testing cleaner.